

Trial 001: Numerical-GP Wildfire-Spread Prediction via Universal-Kriging Initialisation and Ensemble Pushforward of the Rothermel Level-Set Operator

1 Governing Physics

Let $\phi^n(\mathbf{x})$ be the level-set field on the grid $\mathbf{x} = (x, y)$ at time $t_n = n \Delta t$, with $\phi < 0$ burned, $\phi > 0$ unburned, and the fire front at $\phi = 0$. The simulator (`simulation.ipynb`) evolves the Rothermel-driven level set

$$\phi_t = \mathcal{N}[\phi] := -R_f(\phi) |\nabla \phi| + \nu \Delta x \nabla^2 \phi, \quad (1)$$

where $\nu = 0.4$ is the (numerical) viscosity coefficient and the $\nu \Delta x$ prefactor is taken verbatim from the simulator. The rate of spread R_f is the Rothermel (1972) imperial form, decomposed (slope = 0, height field zeroed) as

$$R_f(\phi) = R_0 (1 + \phi_W(\mathbf{x})), \quad R_0 = \frac{I_R \xi}{\rho_b \varepsilon_h Q_{ig}}, \quad (2)$$

with reaction intensity I_R , propagating-flux ratio ξ , oven-dry bulk density ρ_b , effective heating number ε_h , and heat of preignition Q_{ig} . The wind factor couples the front *normal* $\hat{\mathbf{n}} = \nabla \phi / |\nabla \phi|$ to the midflame wind $\mathbf{w}$:

$$\phi_W(\mathbf{x}) = C U(\mathbf{x})^B (\beta / \beta_{op})^{-E}, \quad U(\mathbf{x}) = \max(0, \mathbf{w} \cdot \hat{\mathbf{n}}(\mathbf{x})), \quad (3)$$

so R_f is largest where the front normal aligns with the wind (downwind, NE here) and falls to R_0 where $U = 0$. The clamp $U \geq 0$ avoids fractional powers of negatives (simulator behaviour). Eq. (3) makes (1) *nonlinear and non-local in $\nabla \phi$* .

2 Method

2.1 Prior placement

We place the GP prior on the *unknown initial field* ϕ^0 (the only latent quantity; the operator $\mathcal{N}$ is known). To keep the prior general and free of any ignition-geometry assumption, we use universal kriging: a generic low-order trend mean plus a stationary RBF residual,

$$\phi^0(\mathbf{x}) \sim \mathcal{GP}(m_{\beta}(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad m_{\beta}(\mathbf{x}) = \mathbf{p}(\mathbf{x})^\top \boldsymbol{\beta}, \quad k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 e^{-\|\mathbf{x} - \mathbf{x}'\|^2 / 2\ell^2}. \quad (4)$$

Here $\mathbf{p}(\mathbf{x}) = [1, u, v, u^2, v^2, uv]$ with $u = (x - 100)/100$, $v = (y - 100)/100$ (normalised by the public domain half-width, not the unknown fire location). **The quadratic trend is a smooth bowl; it cannot represent the square / signed-distance front. All front geometry is**

Table 1: Constants and parameters (from `simulation.ipynb`; see `gp/UNITS.md`).

Symbol	Value	Units	Description
grid	200×200	cells	$5 \text{ km} \times 5 \text{ km}$
Δx	82.021	ft (= 25 m)	grid spacing
Δt	1	min	implicit Euler step
ν	0.4	–	viscosity coefficient
$\mathbf{w}$	(100, 100)	ft/min	midflame wind ($ \mathbf{w} = 141.4$)
R_0	3.8065	ft/min	base ROS (no wind)
$R_f^{\max}$	9.6834	ft/min	full-wind ROS ($U = \mathbf{w} $)
I_R	730.93	–	reaction intensity
ξ	0.05775	–	propagating-flux ratio
ρ_b	0.0340	–	ovendry bulk density
ε_h	0.96134	–	effective heating number
Q_{ig}	339.28	–	heat of preignition
C, B	$5.42 \times 10^{-5}, 2.0712$	–	wind-factor constants
$(\beta/\beta_{op})^{-E}$	1.0000	–	wind-factor packing term

supplied by the RBF residual conditioned on data. We realise both blocks as one Bayesian linear model with feature map $\Phi(\mathbf{x}) = [\mathbf{p}(\mathbf{x})^\top, \mathbf{z}(\mathbf{x})^\top]^\top$, where the random Fourier features

$$z_j(\mathbf{x}) = \sqrt{\frac{2}{M}} \sigma_f \cos(\boldsymbol{\omega}_j^\top \mathbf{x} + b_j), \quad \boldsymbol{\omega}_j \sim \mathcal{N}(\mathbf{0}, \ell^{-2} I), \quad b_j \sim U[0, 2\pi], \quad (5)$$

satisfy $\sigma_f^2 \mathbb{E}[\mathbf{z}(\mathbf{x})^\top \mathbf{z}(\mathbf{x}')] \rightarrow k(\mathbf{x}, \mathbf{x}')$ as $M \rightarrow \infty$. With weight prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Lambda)$, $\Lambda = \text{blkdiag}(\tau^2 I_6, I_M)$ (flat on the trend, $\tau = 10^3$), the field is $\phi^0(\mathbf{x}) = \Phi(\mathbf{x})^\top \mathbf{w}$.

2.2 Euler discretisation and numerical GP construction

The embedded time-stepper is forward Euler on (1) ($\Delta t = 1$), defining the (nonlinear) Euler map $\mathcal{E}$:

$$\phi^n = \mathcal{E}[\phi^{n-1}] := \phi^{n-1} + \Delta t \mathcal{N}[\phi^{n-1}] = \phi^{n-1} + \Delta t (-R_f^{n-1} |\nabla \phi^{n-1}| + \nu \Delta x \nabla^2 \phi^{n-1}). \quad (6)$$

Following the Numerical GP construction, linearising $\mathcal{E}$ about the current mean m_{n-1} gives the Jacobian $J_{n-1} = I + \Delta t \mathcal{N}'[m_{n-1}]$ and the joint multi-output GP for two consecutive steps,

$$\begin{bmatrix} \phi^n \\ \phi^{n-1} \end{bmatrix} \sim \mathcal{GP} \left(\begin{bmatrix} \mathcal{E}[m_{n-1}] \\ m_{n-1} \end{bmatrix}, \begin{bmatrix} J_{n-1} K_{n-1} J_{n-1}^\top + Q & J_{n-1} K_{n-1} \\ K_{n-1} J_{n-1}^\top & K_{n-1} \end{bmatrix} \right), \quad (7)$$

i.e. the cross-covariance is the operator applied to the previous kernel, $k^{n,n-1}(\mathbf{x}, \mathbf{x}') = (J_{n-1} \cdot k^{n-1,n-1})(\mathbf{x}, \mathbf{x}')$, and the marginal propagates as $m_n = \mathcal{E}[m_{n-1}]$, $K_n = J_{n-1} K_{n-1} J_{n-1}^\top + Q$ ($Q \succeq 0$ optional process noise; $Q = 0$ here).

Computational realisation (ensemble pushforward). A dense $K_n \in \mathbb{R}^{40000 \times 40000}$ is intractable, and the strong nonlinearity of $|\nabla \phi|$ and (3) makes a single Jacobian linearisation lossy. We therefore propagate the covariance by Monte-Carlo pushforward of the $t = 0$ posterior through the *exact* operator $\mathcal{E}$: draw N posterior samples ϕ_s^0 , advance each by (6), and estimate

$$m_n \approx \frac{1}{N} \sum_{s=1}^N \phi_s^n, \quad K_n \approx \frac{1}{N-1} \sum_{s=1}^N (\phi_s^n - m_n)(\phi_s^n - m_n)^\top, \quad \phi_s^n = \mathcal{E}^n[\phi_s^0]. \quad (8)$$

This is (7) without the first-order truncation: the ensemble carries the GP covariance to every evaluation time and captures curvature/wind nonlinearity that J_{n-1} alone would drop. We report the diagonal $\sigma_n(\mathbf{x}) = \sqrt{\text{diag } K_n}$ as predictive uncertainty.

2.3 Inference from sparse observations

With observation model $y_i = \phi^0(\mathbf{x}_i) + \eta_i$, $\eta_i \sim \mathcal{N}(0, \sigma_n^2)$, $i = 1 \dots 50$, the weight posterior is exact Bayesian linear regression,

$$\mathbf{w} \mid \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_w, \Sigma_w), \quad \Sigma_w = A^{-1}, \quad A = \sigma_n^{-2} \boldsymbol{\Phi}_X^\top \boldsymbol{\Phi}_X + \Lambda^{-1}, \quad \boldsymbol{\mu}_w = \sigma_n^{-2} A^{-1} \boldsymbol{\Phi}_X^\top \mathbf{y}, \quad (9)$$

where $\boldsymbol{\Phi}_X \in \mathbb{R}^{50 \times (6+M)}$ stacks the features at the observation cells. The posterior field mean is $m_0(\mathbf{x}) = \boldsymbol{\Phi}(\mathbf{x})^\top \boldsymbol{\mu}_w$ and posterior IC draws are $\phi_s^0 = \boldsymbol{\Phi} \mathbf{w}_s$, $\mathbf{w}_s = \boldsymbol{\mu}_w + R^{-\top} \boldsymbol{\eta}_s$ with $A = R R^\top$, $\boldsymbol{\eta}_s \sim \mathcal{N}(0, I)$. The lengthscale ℓ is selected by maximising the exact-RBF log marginal likelihood of the *detrended* observations over $\ell \in \{8, \dots, 24\}$ (data-only; no ground truth), giving $\ell = 16$ cells.

2.4 Algorithm

1. Sample 50 observation values from ϕ^0 in the central window $[50, 150]^2$ (seed 42).
2. Select ℓ by detrended marginal likelihood; form the BLR posterior (9) from the 50 values only.
3. Draw $N = 128$ posterior initial fields ϕ_s^0 .
4. Advance each ϕ_s^0 by the embedded Euler map (6) for 300 steps, recording $t \in \{50, 100, 150, 200, 250, 300\}$.
5. Report ensemble mean (8) as the prediction and ensemble std as uncertainty; compare to the simulator ground truth.

3 Assumptions and approximations

- **Known operator, faithful re-implementation.** The embedded operator (6)–(3) is mathematically identical to the simulator’s Rothermel level-set Euler step but is a separate implementation (no shared function). Consequently forecast error is dominated by the *propagated* $t = 0$ *reconstruction error*, reported separately from the reconstruction error itself (Sec. 5).
- **Universal-kriging mean.** The quadratic trend is a generic smooth bowl, not an ignition-shape prior. In the unobserved far field the posterior reverts to this trend, which overshoots the true signed distance; hence *full-grid* RMSE is dominated by the far field, while the fire (which never leaves $[80, 140]^2$) is governed by the central window.
- **Covariance via ensemble pushforward** (8), not a dense kernel; first-order term in (7) is superseded by the exact nonlinear pushforward. RFF ($M = 512$) approximates the RBF kernel.
- **No reinitialisation** of ϕ on *either* side (symmetric): the simulator never redistances, so $|\nabla \phi|$ drifts from 1 identically in prediction and ground truth.
- **Boundaries ignored** (infinite-plane assumption); verified that the front stays interior. Wind clamp $U \geq 0$ as in the simulator. Euler only, $\Delta t = 1$ min.

4 Experimental setup

Seed = 42; grid 200×200 ; $\Delta x = 82.021$ ft; $\Delta t = 1$ min; $\nu = 0.4$; wind (100, 100) ft/min. Observations: 50 cells drawn uniformly from $[50, 150]^2$. GP: $\ell = 16$ cells, $\sigma_f = 13.44$, $\sigma_n = 0.1$, $\tau = 10^3$, $M = 512$ random Fourier features. Ensemble $N = 128$. Evaluation $t \in \{50, 100, 150, 200, 250, 300\}$. **Code paths (disjoint):** ground truth = `simulator_groundtruth_ground_truth_step`; prediction = `numerical_gp_embedded_levelset_step`. They share no forward-stepping function (asserted at run time).

5 Results

Table 2 separates the $t = 0$ reconstruction error (GP posterior vs. true ϕ^0) from the $t > 0$ forecast error. The fire front is tracked to 1.3–1.5 cells (≈ 110 –120 ft, i.e. 1.3–1.5 Δx) throughout, while the field RMSE in the observed window stays ≈ 4.4 –4.7. The Hausdorff front distance (≈ 4.2 cells) reflects the persistent mismatch at the sharp ignition corners that the RBF reconstruction rounds off (Fig. 1): the test is not circular — the error is real, distributed, and concentrated where reconstruction is hardest.

Table 2: Metrics. Row $t = 0$ is reconstruction (separate from forecast). RMSE/MAE in ϕ (cells); front distances in cells (mean) and ft; $\bar{\sigma}_{\text{band}}$ is ensemble std in the front band $|\phi_{\text{gt}}| \leq 4$; last column is the GP-ablation (trend-only IC) window RMSE.

t	RMSE _{win} (ϕ)	RMSE _{full} (ϕ)	front mean (cells)	front mean (ft)	Hausdorff (cells)	$\bar{\sigma}_{\text{band}}$ (cells)	abl. RMSE _{win} (no front)
0 (recon)	4.205	11.567	1.45	119	–	0.183	–
50	4.428	12.088	1.41	116	4.41	0.183	7.213
100	4.526	11.996	1.41	115	4.38	0.181	7.949
150	4.554	12.137	1.37	113	4.33	0.181	8.831
200	4.552	12.407	1.36	112	4.29	0.182	9.825
250	4.581	12.763	1.33	109	4.18	0.191	10.898
300	4.696	13.284	1.33	109	4.23	0.256	12.019

GP is load-bearing (two diagnostics). (i) *Uncertainty propagation.* The ensemble std in the front band is non-zero at every time and grows 0.183 $\rightarrow$ 0.256 cells of ϕ from $t = 0$ to $t = 300$ (Fig. 4, left) — the nonlinear operator amplifies the inferred initial-state uncertainty, so this is not a mean-only rollout. (The absolute predictive variance is far larger, > 60 cells, in the unobserved far field.) (ii) *Ablation.* Removing the GP residual (quadratic trend only) yields an IC with $\min \phi = 14.8 > 0$: **no fire front exists at all** (0 burned cells at every time), and the window RMSE degrades to 7.2 $\rightarrow$ 12.0 (Fig. 4, right). The entire fire geometry is supplied by the GP residual conditioned on the 50 values; the result changes drastically when the GP is removed.

Circularity / leakage self-check (`self_check.txt`): (a) predictions come from `embedded_levelset_step`, distinct from the simulator’s `ground_truth_step` (asserted `True`); (b) inference used only the 50 observation values (range $[-3.0, 47.5]$); the GP never reads the true field or its parameters; (c) the GP is non-degenerate (front-band std 0.18–0.26 > 0 ; reconstruction window RMSE 4.2 > 0 , so the front is genuinely inferred, not leaked). No spurious burned cells appear outside the window.

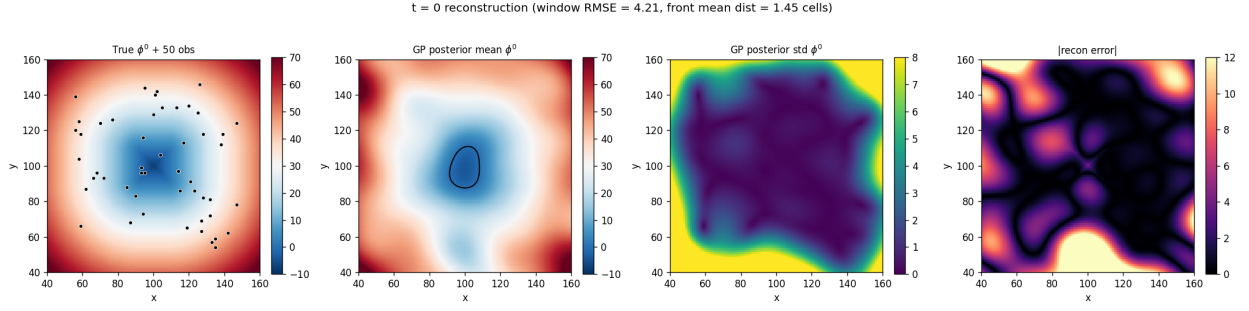


Figure 1: $t = 0$. True ϕ^0 with the 50 sampled observations (black), GP posterior mean (note the rounded reconstruction of the sharp square front), posterior std (low where observed, high at unobserved corners), and $|\text{reconstruction error}|$.

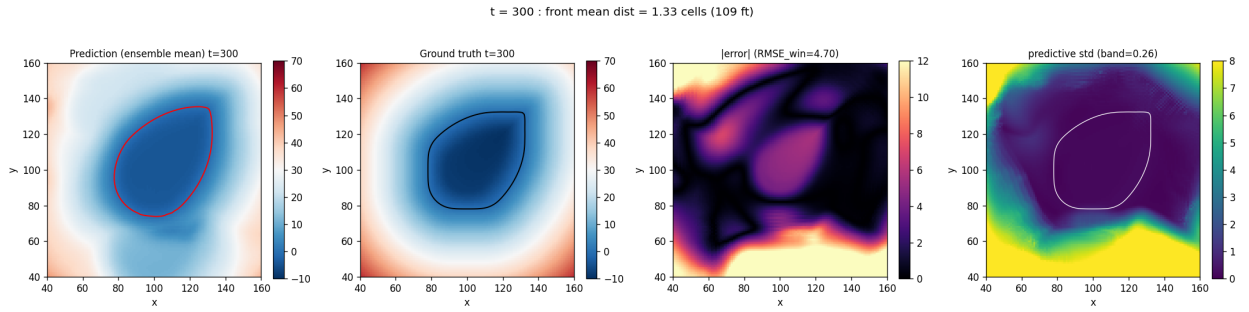


Figure 2: $t = 300$. Prediction (ensemble mean, red contour) vs. ground truth (black), absolute error, and predictive std with the GT front overlaid (white). Fronts agree to 1.33 cells mean; shape differences persist at the wind-driven tip.

6 Sources

- M. Raissi, P. Perdikaris, G. E. Karniadakis, *Numerical Gaussian Processes for Time-dependent and Non-linear Partial Differential Equations* — joint multi-output GP / Euler-embedded kernel construction (7).
- R. C. Rothermel (1972), *A Mathematical Model for Predicting Fire Spread* — rate-of-spread (2)–(3) (gp/docs/).
- D. Muñoz-Esparza et al. (2018), *An Accurate Fire-Spread Algorithm in the WRF Model Using the Level-Set Method* — level-set form (1) (gp/docs/).
- `simulation.ipynb` — source of truth for constants, operators, and ground truth; `gp/UNITS.md` — units ($\Delta t = 1$ min, $\Delta x = 82.021$ ft).

7 Notes

- **Worked:** sparse central inference + identical physics gives a front tracked to ~ 1.4 cells; the level-set speed $F = R_f$ is independent of $|\nabla\phi|$, so once the front is located it stays located (the slight front-distance *decrease* over time is curvature/viscosity smoothing of the rounded prediction toward the GT).

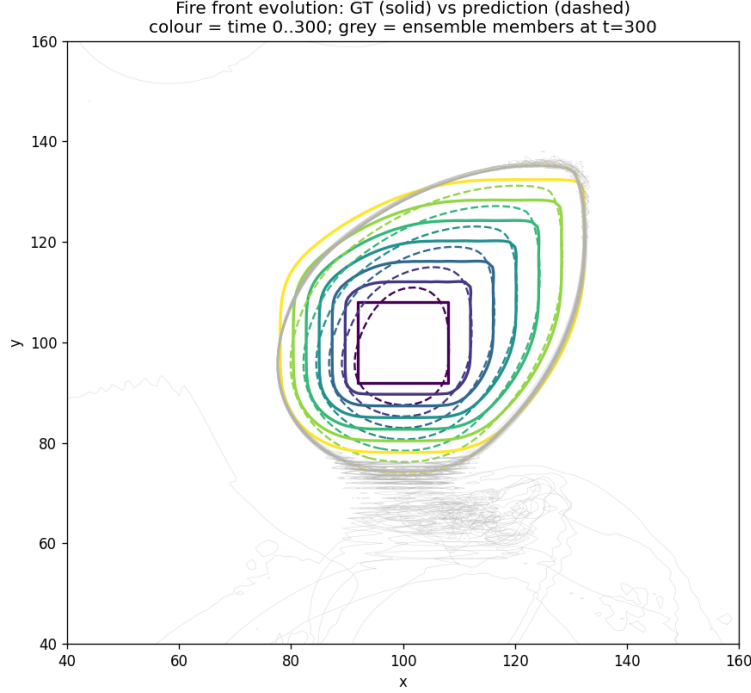


Figure 3: Fire-front evolution, GT (solid) vs. prediction (dashed), colour-coded in time $0 \rightarrow 300$. The $t = 0$ GT front is the sharp square; the prediction starts rounded yet tracks the teardrop spread. Grey curves are 40 ensemble members at $t = 300$ (uncertainty band).

- **Limitations:** (1) far-field reconstruction reverts to the quadratic trend (full-grid RMSE inflated; fire unaffected); (2) front-band predictive std is modest (0.18–0.26 cells) because 50 central observations strongly constrain the front; (3) the operator is identical by design, so this trial tests *initial-state inference* + *uncertainty propagation*, not operator learning.
- **Next-trial ideas:** replace the quadratic trend with a monotone-radial-basis far-field model; condition on fewer / noisier observations to enlarge predictive variance; add process noise Q from the Euler truncation; compare ensemble pushforward against the explicit linearised $K_n = J_{n-1}K_{n-1}J_{n-1}^\top$ propagation.

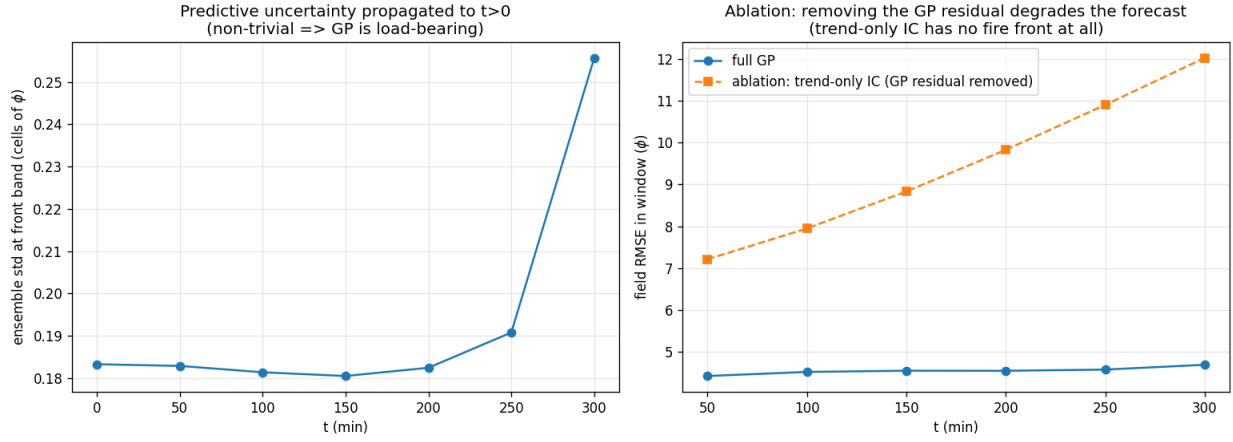


Figure 4: Load-bearing diagnostics. Left: front-band ensemble std grows with time (uncertainty propagated, not collapsed). Right: ablation — removing the GP residual (trend-only IC, which has no front) degrades window RMSE from ≈ 4.5 to 7–12.